

Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_modular_operator_compression_pressure.
- Best held-out TSPLIB gap: phase7_modular_operator_compression_pressure.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260520_024159_i
- started_at_local: 2026-05-20 02:41:59
- finished_at_local: 2026-05-20 03:15:15
- duration_hhmm: 00:33
- duration_seconds: 1996.724
- seed_offset: 8000
- replicate_label: i
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.213387	0.064916	0.159398	0.893241	0.56	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235058	0.054968	0.121317	0.806663	0.4375	False	0.002824	0.016003
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235058	0.054968	0.121317	0.880523	0.445	False	0.004089	-0.00556
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.235667	0.054968	0.121542	0.84406	0.4425	False	0.003334	0.235658

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.176029	0.04601	0.093911	0.0	0.445	False	-0.038703	0.981692
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.236585	0.064041	0.12761	0.81	0.4425	False	0.002996	0.019176

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.213387, family holdout 0.064916, combined 0.159398.
- Accepted novelty 0.893241, complexity 0.56, and adaptation efficiency -0.027882.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.806663, complexity 0.4375, and adaptation efficiency 0.007377.
- Last-epoch runtime 71.45605 ms and distance evaluations 5179.0.
- Validation: surviving False, transplant delta 0.002824, positive scaffolds 0, Pareto runtime inflation 0.016003.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": 0.016003, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.880523, complexity 0.445, and adaptation efficiency 0.006758.
- Last-epoch runtime 60.68015 ms and distance evaluations 3810.75.

- Validation: surviving False, transplant delta 0.004089, positive scaffolds 0, Pareto runtime inflation -0.00556.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": -0.00556, "same_gap_faster": true}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.84406, complexity 0.4425, and adaptation efficiency 0.009631.
- Last-epoch runtime 67.84825 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.003334, positive scaffolds 1, Pareto runtime inflation 0.235658.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.232518, "gap_delta": 0.000224, "runtime_inflation": 0.235658, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.176029, family holdout 0.04601, combined 0.093911.
- Accepted novelty 0.0, complexity 0.445, and adaptation efficiency 0.0.
- Last-epoch runtime 50.604475 ms and distance evaluations 4998.5.
- Validation: surviving False, transplant delta -0.038703, positive scaffolds 4, Pareto runtime inflation 0.981692.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 11.735255, "gap_delta": -0.02318, "runtime_inflation": 0.981692, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.236585, family holdout 0.064041, combined 0.12761.
- Accepted novelty 0.81, complexity 0.4425, and adaptation efficiency 0.006923.
- Last-epoch runtime 127.286725 ms and distance evaluations 7395.75.
- Validation: surviving False, transplant delta 0.002996, positive scaffolds 0, Pareto runtime inflation 0.019176.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 0.003718, "gap_delta": 0.006293, "runtime_inflation": 0.019176, "same_gap_faster": false}

Judge Appendix

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Review Summary of TSP Modular-Operator Discovery Suite

Key Evaluation Criteria Recap

- ****Main Transfer Evidence:**** Held-out TSPLIB gap and family holdout gap.
- ****Operator Interest:**** Only if validation survives transplant and ablation tests.
- ****Claims:**** Prefer reusable operator structure over full-solver performance.

- **Algorithm Discovery:** Only if operator survives scaffold and family holdout tests.

Overall Outcome

- **No surviving candidates** under the stringent criteria.
- **No operator shows validated benefit** after ablation and transplant tests on held-out sets.

Condition Highlights and Interpretation

Condition Name	Operator Type	Final Family Gap	Final TSPLI B Gap	Transfer Gap	Ablation Support	Transplant Findings	Surviving Candidate?
phase7_baseline_heuristic_only	n/a	0.05497	0.23506	0.12132	n/a	n/a	No
phase7_full_solver_evolution	n/a	0.06492	0.21339	0.15940	n/a	n/a	No
phase7_modular_operator_evolution	perturbation	0.05497	0.23506	0.12132	Ablation negative (disabled no effect)	No improvement on scaffold transplantation; fails on clustered_local_search	No
phase7_modular_operator_random_replay	perturbation	0.05497	0.23506	0.12132	Ablation negative (disabled no effect)	No positive scaffold transplant; no family gain	No
phase7_modular_operator_diversity_residual_replay	candidate_pruner	0.05497	0.23567	0.12154	Ablation negative (disabling no hurt)	One positive scaffold transplant but no family gap improvement; high runtime inflation	No
phase7_modular_operator_compression_pressure	scaffold_selector	0.04601 (best)	0.17603 (best)	0.09391	Ablation shows disabling hurts (gap +0.0219), some variants better	Positive transplant mean gap delta (-0.0387); mixed family results with 2 positive and 2 severe regressions; runtime inflation high (~98%)	No
phase7_modular_operator_pareto_selection	candidate_ranker	0.06404	0.23659	0.12761	Ablation negative (disabled improves gap)	No positive scaffold transplant; no family gain	No

Conservative Interpretation

1. **Best Transfer Performance and Holistic Gap:**

The `*phase7_modular_operator_compression_pressure*` condition shows the best held-out TSPLIB gap (0.176 vs baseline 0.235) and the best family gap (0.046), indicating some transfer potential.

2. **Ablation and Operator Effect:**

- This compression pressure operator (a scaffold selector) is the only operator where disabling it increases gap substantially (+0.0219), suggesting it has some positive internal effect.
- Other operators (perturbations, candidate rankers/pruners) show no loss when disabled, indicating no validated beneficial operator effect.

3. **Transplant Test:**

- Compression pressure operator shows a **negative mean transplant gap delta (-0.0387)**, meaning it improved performance on transplanted scaffolds in aggregate, with 4 positive scaffolds out of 5 tested.
- However, it does not transplant cleanly into some scaffolds (e.g., fails for `cheapest_insertion_2opt`), and shows regressions on some families (`nearest_neighbor_trap_tsp`, `uniform_euclidean`).

4. **Family Holdout Signal:**

- Compression pressure operator has 2 positive families (`heldout_tsplib` and `two_cluster_bottleneck_tsp`) with improved gaps, but also 2 with severe regressions.
- Other operators show no meaningful family improvement.

5. **Runtime and Complexity:**

- Compression pressure operator incurs high runtime inflation (~98%), which diminishes practical value despite gap improvement.
- Other operators have much lower runtime impact but lack validated improvements.

6. **Operator Survival:**

- No operator fulfills all criteria: survival under scaffold and family holdout tests AND positive ablation AND positive transplant validation.
- Thus, **no modular operator qualifies as a surviving reusable discovery or validated new algorithmic operator.**

Specific Notes on Operator Types

- **Perturbation Operators (double-bridge escalation):**

Fail ablation (disabling has zero gap impact), no family transfer gain, do not transplant well—suggesting no validated reusable effect despite novelty.

- **Candidate Pruner / Ranker Operators:**

No validated improvement; ablations often show better gaps when operator is disabled, no positive family or transplant validation.

- **Scaffold Selector (compression pressure):**

Unique among tested operators: shows a reproducible positive transfer pattern, ablation shows dependency, and transplant improvements, but also runtime cost and inconsistent family effects prevent full validation.

Final Conclusion

- ****No modular operator discovered in this suite meets the rigorous criteria for validated reusable operator structure.****
- ****Best candidate is a deterministic scaffold selector**** that improves held-out TSPLIB performance and transplant mean gap but suffers from high runtime cost and some family regressions.
- ****No algorithm discovery claim is warranted.****
- ****No modular operator survives ablation and transplant validations simultaneously with positive holdout gains.****

Summary Table

Criterion	Result
Best held-out TSPLIB gap	0.17603 (compression pressure operator)
Best family holdout gap	0.04601 (compression pressure operator)
Survival of ablation test	Only compression pressure operator shows gap worsening when disabled; others no effect or better without operator
Survival of transplant	Compression pressure operator shows transplant gap improvements on most scaffolds; others do not
Family holdout validation	Mixed for compression pressure (2 positive, 2 regression); none positive for others
Runtime inflation	Very high (~98%) for compression pressure, low for others
Surviving candidates	None
Algorithm discovery	Not supported

Recommendations

- Further reduce runtime overhead for promising scaffold selector to improve practical viability.
- Explore operator variants that preserve or improve ablation signal and broaden family positive signal.
- Do not claim discovery of new reusable operators nor new full solvers based on current results.
- Maintain focus on held-out gap, ablation validation, and transplant robustness for future candidates.

****End of conservative analysis.****